Experiment Title: Helpful Utilities in Linux Terminal

Topic: Practical Everyday Utilities for Terminal Management, System Info, and Environment

1. Objective

To practice everyday Linux terminal utilities:

- Check user identity and system name (**`whoami`**, **`who`**, **`hostname`**).
- Display text, timestamps, and insert delays (**`echo`**, **`date`**, **`sleep`**).
- Clear terminal screen (**`clear`**) and read command manuals (**`man`**).
- View past commands (**`history`**).
- Create command shortcuts (**`alias`**) and view environment variables (**`env`**, **`printenv`**).

2. Theoretical Background

Linux provides essential utilities for identity, screen control, documentation, history, aliases, and environment variables.

2.1 Commands Practiced in This Lab

Command	Syntax	Simple Meaning / Purpose	Example
`whoami`	`whoami`	Print current username.	`whoami`
`who`	`who`	List logged-in users.	`who`
`hostname`	`hostname`	Print system network name.	`hostname`
`date`	`date`	Display current date and time.	`date`
`echo`	`echo <text>`	Print text to terminal.	`echo "Hello"`
`sleep`	`sleep <N>`	Pause execution for N seconds.	`sleep 2`
`clear`	`clear`	Clear terminal screen.	`clear`
`man`	`man <cmd>`	View manual page for a command.	`man echo`
`history`	`history`	List previously run commands.	`history`
`alias`	`alias name='cmd'`	Create shortcut for command.	`alias ll='ls -lh'`
`env`	`env`	Display environment variables.	`env`
`printenv`	`printenv [var]`	Print value of environment variable.	`printenv SHELL`

2.2 Grouping Utilities by Everyday Function

- **Identity:** **`whoami`**, **`who`**, **`hostname`**, and **`date`** display user and system status.
- **Control & Help:** **`clear`** clears screen, **`echo`** prints text, **`sleep`** pauses, **`man`** shows documentation.
- **Workflow:** **`alias`** creates shortcuts, **`history`** lists past commands, **`env`**/**`printenv`** check environment variables.

3. Equipment / Software Requirements

Item	Details	Item	Details
Operating System	Linux (Ubuntu 24.04 LTS)	**Host Machine**	`firefly`
User Account	`ratul`	**Command Shell**	GNU Bash 5.2
Terminal Emulator	Linux Terminal (Kitty / Ghostty)	**Core Utilities**	GNU Coreutils, Bash Builtins, Man-db

4. Procedure / Steps

1. Check user identity with **whoami** and system name with **hostname**.
2. List logged-in users with **who** and display system time with **date**.
3. Print message with **echo** and test delay with **sleep 2**.
4. Clear terminal screen using **clear**.
5. View manual page using **man echo**.
6. Create and test command shortcut using **alias ll='ls -lh'**.
7. Inspect environment variables using **printenv** and **env**.
8. View command history using **history**.

5. Result / Output Demonstration

Commands run on Ubuntu under user `ratul`:

5.1 Session 1: Identity, System Time, Output Display, and Documentation

ratul@firefly: ~ — bash

```
ratul@firefly:~$ hostname && whoami
firefly
ratul
ratul@firefly:~$ who
ratul   pts/0      2026-10-08 09:15 (:0)
ratul@firefly:~$ date
Thu Oct 8 10:30:15 UTC 2026
ratul@firefly:~$ echo "Testing delay..." && sleep 2 && echo "Ready!"
Testing delay...
Ready!
ratul@firefly:~$ clear
[Screen cleared successfully]
ratul@firefly:~$ man echo | head -n 4
ECHO(1)                    User Commands                    ECHO(1)
NAME
    echo - display a line of text
ratul@firefly:~$ _
```

Explanation of Session 1 Results:

- **hostname** & **whoami**: Displayed machine name (`firefly`) and username (`ratul`).
- **who**: Showed active session on terminal `pts/0`.
- **date**: Displayed system date, time, and timezone.
- **echo** & **sleep**: Printed text and paused execution for 2 seconds.
- **clear** & **man**: Cleared terminal screen and displayed manual documentation.

5.2 Session 2: Command Shortcuts (Alias), Environment Variables, and History

ratul@firefly: ~ — bash

```
ratul@firefly:~$ alias ll="ls -lh"
ratul@firefly:~$ ll
total 8.0K
drwxr-xr-x 2 ratul users 4.0K Oct  8 09:30 Documents
-rw-r--r- 1 ratul users 1.2K Oct  8 10:12 notes.txt
ratul@firefly:~$ printenv USER SHELL
ratul
/bin/bash
ratul@firefly:~$ env | grep -E "USER|HOSTNAME|HOME"
HOME=/home/ratul
HOSTNAME=firefly
USER=ratul
ratul@firefly:~$ history | tail -n 6
  42  hostname && whoami
  43  date
  44  echo "Testing delay..." && sleep 2
  45  alias ll="ls -lh"
  46  printenv USER SHELL
  47  history | tail -n 6
ratul@firefly:~$ _
```

Explanation of Session 2 Results:

- **`alias ll="ls -lh"`**: Created shortcut `ll` for detailed file listing.
- **`printenv`**: Displayed values of `USER` and `SHELL`.
- **`env | grep`**: Filtered and displayed specific environment variables.
- **`history`**: Displayed numbered list of executed commands.

6. Conclusion

Key takeaways:

1. **`whoami`**, **`who`**, **`hostname`**, and **`date`** inspect user and machine identity.
2. **`echo`**, **`sleep`**, and **`clear`** control output and screen flow; **`man`** provides documentation.
3. **`alias`** saves keystrokes with shortcuts; **`history`** recalls past commands.
4. **`env`** and **`printenv`** inspect system environment variables.